

Worksheet No. - 03

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Aim/Overview of the practical: (A). To interface the MFRC522 RFID reader module with an Arduino Uno and to scan/read the Unique Identification (UID) number of an RFID tag using serial communication (SPI protocol), and to display the UID on the Serial Monitor.

APPARATUS / COMPONENTS REQUIRED

S.No	Component / Apparatus	Quantity	Specification
1	Arduino Uno Board	1	ATmega328P based
2	MFRC522 RFID Reader Module	1	13.56 MHz, SPI interface
3	RFID Tag / Card	1 or 2	Mifare 1K, 13.56 MHz
4	Jumper Wires (Male-Female)	7	-
5	USB Cable (Type A to B)	1	For Arduino-PC connection
6	Computer with Arduino IDE	1	Version 1.8.x or later
7	Breadboard (optional)	1	For stable connections

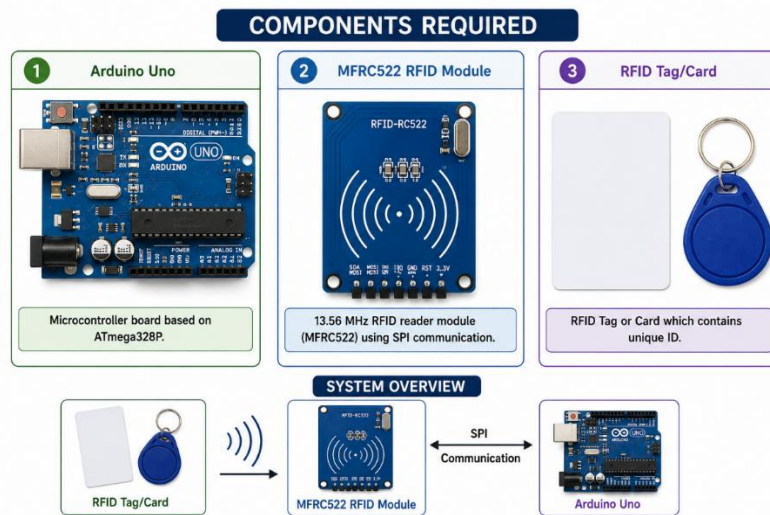
THEORY

RFID (Radio Frequency Identification) is a wireless technology used to identify and track tags attached to objects using radio waves. An RFID system consists of two main parts — an RFID reader (also called a PICC reader) and an RFID tag (transponder). Each tag contains a unique 4-byte or 7-byte identification number called the UID (Unique Identifier), which is permanently programmed into the tag during manufacture.

The MFRC522 is a highly integrated 13.56 MHz proximity RFID reader/writer IC manufactured by NXP Semiconductors. It supports the ISO/IEC 14443 A/MIFARE standard and communicates with a microcontroller using the SPI (Serial Peripheral Interface) protocol. When an RFID tag is brought within the reader's magnetic field (typically 2-5 cm), the tag draws energy from the field, activates its internal circuit, and transmits its UID back to the reader through load modulation.

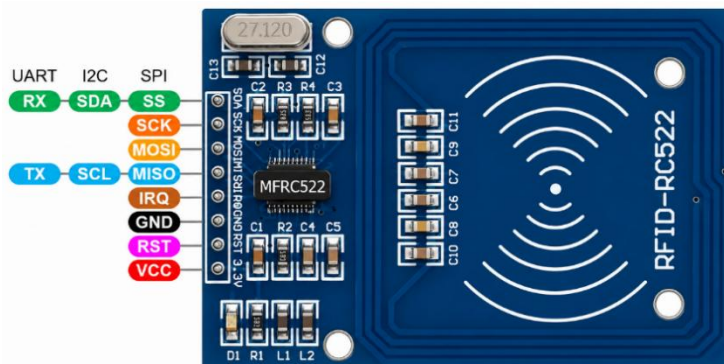
An RFID system mainly consists of three components:

- RFID Tag (Transponder)
- RFID Reader (Interrogator)
- Microcontroller or Processing Unit



The RC522 is an RFID (Radio Frequency Identification) reader/writer module commonly used with Arduino projects for contactless identification. The RC522 RFID Module is a widely used RFID reader and writer module designed for short-range wireless communication. It is commonly interfaced with microcontrollers such as the Arduino Uno to build security systems, attendance systems, smart cards, and automation projects.

The module is based on the MFRC522 integrated circuit developed by NXP Semiconductors. This chip handles all RFID communication tasks such as generating radio frequency signals, detecting tags, authentication, and data exchange.



Technical Specifications

Feature	Description
Module	RC522 / MFRC522
Technology	RFID / NFC-compatible 13.56 MHz cards and tags
Operating voltage	3.3 V
Communication	SPI, commonly used with Arduino
Frequency	13.56 MHz
Typical range	A few centimeters
Applications	Access control, attendance, identification, smart locks
Arduino compatibility	Arduino Uno, Nano, Mega, etc.

Operating Principle

The RC522 module works at a frequency of 13.56 MHz, which belongs to the High Frequency (HF) RFID band.

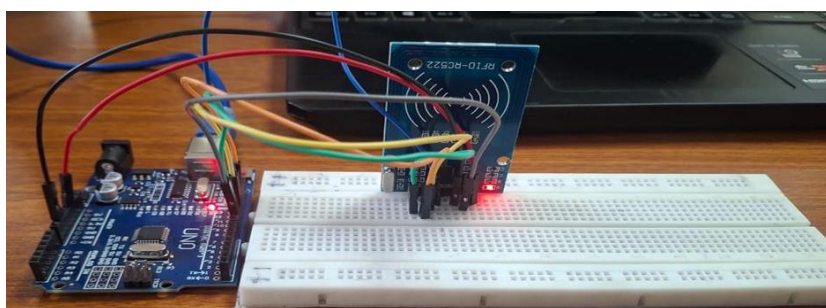
When powered:

- The module generates an electromagnetic (EM) field using its onboard antenna.
- When an RFID card or tag enters this field:
- A passive tag gets powered by the EM waves.
- The tag activates and sends its stored data (usually UID).
- The RC522 receives this data.
- The data is sent to Arduino via SPI communication.

CIRCUIT / PIN CONNECTION DIAGRAM

Pin connection details between the MFRC522 module and Arduino Uno:

MFRC522 Pin	Arduino Uno Pin	Function
SDA (SS)	Digital Pin 10	Slave Select (SPI)
SCK	Digital Pin 13	Serial Clock (SPI)
MOSI	Digital Pin 11	Master Out Slave In (SPI)
MISO	Digital Pin 12	Master In Slave Out (SPI)
IRQ	Not Connected	Interrupt (unused)
GND	GND	Common Ground
RST	Digital Pin 9	Reset
3.3V	3.3V	Power Supply



PROCEDURE

1. Connect the MFRC522 module to the Arduino Uno as per the pin connection table given above.
2. Connect the Arduino Uno to the computer using a USB cable.
3. Open the Arduino IDE on the computer.
4. Go to Sketch → Include Library → Manage Libraries, search for "MFRC522" (by GithubCommunity / miguelbalboa) and install it.
5. Select the correct Board (Arduino Uno) and Port from the Tools menu.
 - a. Type or copy the program given below into the Arduino IDE editor.
6. Verify (compile) the program to check for errors.
7. Upload the program to the Arduino Uno.
8. Open the Serial Monitor (Ctrl+Shift+M) and set the baud rate to 9600.
9. Bring an RFID tag/card close to the MFRC522 antenna (within 2-3 cm).
10. Observe the UID and tag type printed on the Serial Monitor.
11. Repeat the process with different RFID tags and record the observations.

PROGRAM / ARDUINO CODE

```
#include <SPI.h>
#include <MFRC522.h>

#define RST_PIN 9
#define SS_PIN 10
#define buzzer 8

MFRC522 mfrc522(SS_PIN, RST_PIN);

byte accessUID[4] = {0x20, 0x4F, 0x78, 0xA2};
// byte accessUID1[4] = {0x19, 0x3A, 0x75, 0xE4};

// byte accessUID[4] = {0x1F, 0x0A, 0xB0, 0xDD};
// byte accessUID1[4] = {0x4B, 0x47, 0x26, 0x1B};

void setup()
{
  //playMelody();
  Serial.begin(9600);

  pinMode(buzzer,OUTPUT);

  // Serial.begin(9600);
  while (!Serial);
  SPI.begin();
  mfrc522.PCD_Init();
```

```

    delay(4);
    mfrc522.PCD_DumpVersionToSerial();
    //Serial.println(F("Scan PICC to see UID, SAK, type, and data blocks..."));
  }

void loop()
{
  if (mfrc522.PICC_IsNewCardPresent() && mfrc522.PICC_ReadCardSerial())
  {
    //Serial.println("button val = " );
    // Serial.println(a);

    if(mfrc522.uid.uidByte[0]==accessUID[0]&& mfrc522.uid.uidByte[1]==accessUID[1]
      &&mfrc522.uid.uidByte[2]==accessUID[2]&&mfrc522.uid.uidByte[3]==accessUID[3] )
    {
      Serial.println("Registered User - PRESENT");
      digitalWrite(buzzer,HIGH);delay(500);
      digitalWrite(buzzer,LOW);delay(500);

    }
    else
    {
      Serial.println("Non-Registered User- contact to your coordinator");
      digitalWrite(buzzer,HIGH);delay(200);
      digitalWrite(buzzer,LOW);delay(200);
      digitalWrite(buzzer,HIGH);delay(200);
      digitalWrite(buzzer,LOW);delay(200);
    }
  }
  // }
  mfrc522.PICC_HaltA();
}
}

```

Important functions used

Function	Purpose
SPI.begin()	Starts SPI communication
mfrc522.PCD_Init()	Initializes RC522
PICC_IsNewCardPresent()	Checks for a new RFID card
PICC_ReadCardSerial()	Reads the card UID
mfrc522.uid.size	Gives the UID length
mfrc522.uid.uidByte[i]	Gets each UID byte
PICC_HaltA()	Stops communication with the card
PCD_StopCrypto1()	Stops encrypted communication
Serial.print()	Displays information on Serial Monitor
delay(100)	Creates a 100 ms delay

OBSERVATION TABLE

S.No	RFID Tag Used	UID Displayed on Serial Monitor	Tag Type	Remarks
1	Scan RFID Card..	04 A7 3B 52	RFID Card	Works

RESULT

The MFRC522 RFID reader module was successfully interfaced with the Arduino Uno using the SPI protocol. The UID of the given RFID tag(s) was successfully scanned and displayed on the Serial Monitor via serial communication.

Sample Output (Serial Monitor):

Scan RFID Card...

Card UID: F2 75 78 0D

PRECAUTIONS

1. Ensure the MFRC522 module is powered with 3.3V only, never 5V.
2. Make all SPI connections (MOSI, MISO, SCK, SS) correctly as per the pin table.
3. Keep the RFID tag within 2-5 cm of the reader's antenna for reliable detection.
4. Avoid keeping the module near metal surfaces, which can interfere with the RF field.
5. Select the correct Board and COM Port in the Arduino IDE before uploading the code.
6. Handle the RFID module carefully as it is sensitive to static discharge.

AIM: (B). To design and implement an RFID-based access control system using Arduino and an MFRC522 RFID reader module, with only two LEDs for indication - a Green LED that glows for the correct (authorized) RFID card and a Red LED that glows for a wrong (unauthorized) RFID card.

APPARATUS / COMPONENTS REQUIRED

S. No.	Component	Quantity	Specification
1	Arduino Uno / Nano	1	ATmega328P based board
2	MFRC522 RFID Reader Module	1	13.56 MHz, SPI interface
3	RFID Tags / Cards	2	13.56 MHz Mifare tags (1 correct, 1 wrong)
4	Green LED	1	5 mm — indicates correct RFID
5	Red LED	1	5 mm — indicates wrong RFID
6	Resistors	2	220 Ω (current limiting for LEDs)
7	Breadboard	1	Standard prototyping board
8	Jumper Wires	As required	Male-to-male / male-to-female
9	USB Cable	1	For Arduino programming
10	Computer with Arduino IDE	1	With MFRC522 library installed

THEORY

RFID Technology

Radio Frequency Identification (RFID) is a wireless technology that uses electromagnetic fields to automatically identify and track tags attached to objects. An RFID system consists of two main parts: a reader (interrogator) and a tag (transponder). The reader emits radio waves; when a tag comes within range, it is powered by the reader's field (in the case of passive tags) and transmits back its stored data, typically a Unique Identifier (UID).

MFRC522 Module

The MFRC522 is a highly integrated reader/writer IC for contactless communication at 13.56 MHz, commonly used with Mifare cards. It communicates with a microcontroller using the SPI (Serial Peripheral Interface) protocol, which requires four main signal lines: SCK (clock), MOSI (master out slave in), MISO (master in slave out), and SDA/SS (slave select), along with RST (reset), GND, and a 3.3V supply.

Access Control Logic

In this project, each RFID tag has a unique, factory-set UID that cannot be duplicated easily. The Arduino continuously polls the MFRC522 for a nearby tag. When a tag is detected, its UID is read and compared against a single pre-stored authorized UID in the program memory. Based on the comparison:

- If the scanned UID matches the authorized UID → the card is 'Correct RFID' → Green LED turns ON.
- If the scanned UID does not match → the card is 'Wrong RFID' → Red LED turns ON.

CIRCUIT DIAGRAM / PIN CONNECTIONS

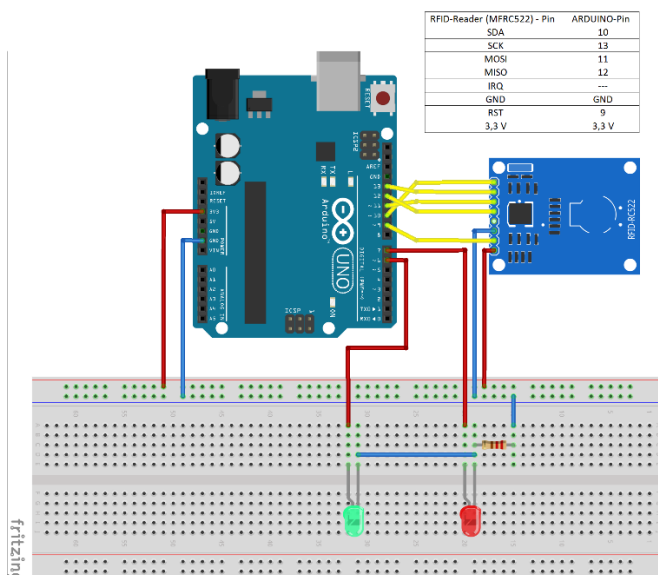
Connect the MFRC522 module to the Arduino Uno as per SPI pin mapping (fixed pins for Uno/Nano):

MFRC522 Pin	Arduino Uno Pin	Function
SDA / SS	D10	Slave Select
SCK	D13	SPI Clock
MOSI	D11	SPI Data Out
MISO	D12	SPI Data In
RST	D9	Reset
3.3V	3.3V	Power Supply
GND	GND	Ground
IRQ	Not Connected	Interrupt

LED connections (only two LEDs used):

Component	Arduino Pin	Connection
Green LED Anode (+)	D6	Through 220 Ω resistor
Green LED Cathode (-)	GND	Ground
Red LED Anode (+)	D7	Through 220 Ω resistor
Red LED Cathode (-)	GND	Ground

- Green LED anode → D6 (via 220 Ω resistor); cathode → GND → glows for CORRECT RFID
- Red LED anode → D7 (via 220 Ω resistor); cathode → GND → glows for WRONG RFID



PROCEDURE

1. Connect the MFRC522 RFID module to the Arduino as per the pin mapping given in pin configuration table.
2. Connect the Green LED to pin D6 and the Red LED to pin D7, each through a 220 Ω resistor, with cathodes to GND.

3. Install the required Arduino library: open Arduino IDE → Sketch → Include Library → Manage Libraries → search 'MFRC522' by GithubCommunity → Install.
4. Upload a UID-reading sketch first and open the Serial Monitor (9600 baud) to scan the correct RFID card and note down its unique UID.
5. Update the 'correctUID' array in the main program (Section 7) with the UID obtained in the previous step.
6. Upload the complete program to the Arduino board.
7. Open the Serial Monitor to observe status messages ('Correct RFID' / 'Wrong RFID').
8. Bring the correct (authorized) RFID card near the MFRC522 antenna and observe that the Green LED turns ON.
9. Bring a different (unauthorized) RFID card near the MFRC522 antenna and observe that the Red LED turns ON.
10. Record observations for multiple trials and tabulate the results.

ARDUINO PROGRAM / SOURCE CODE

```
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 10
#define RST_PIN 9

#define GREEN_LED 7
#define RED_LED 6

MFRC522 rfid(SS_PIN, RST_PIN);

// Replace this UID with your authorized card UID
byte authorizedUID[] = {0xA3, 0x5B, 0x21, 0x1C};

void setup()
{
  Serial.begin(9600);

  SPI.begin();
  rfid.PCD_Init();

  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);

  digitalWrite(GREEN_LED, LOW);
  digitalWrite(RED_LED, LOW);

  Serial.println("RFID Access Control System");
  Serial.println("Scan your RFID card...");
}
```

```
void loop()
{
    // Check if a new RFID card is present
    if (!rfid.PICC_IsNewCardPresent())
    {
        return;
    }

    // Read the RFID card
    if (!rfid.PICC_ReadCardSerial())
    {
        return;
    }

    Serial.print("Card UID: ");

    for (byte i = 0; i < rfid.uid.size; i++)
    {
        Serial.print(rfid.uid.uidByte[i], HEX);
        Serial.print(" ");
    }

    Serial.println();

    // Check whether the card is authorized
    if (checkUID())
    {
        Serial.println("Access Granted!");

        digitalWrite(GREEN_LED, HIGH);
        digitalWrite(RED_LED, LOW);

        delay(3000);

        digitalWrite(GREEN_LED, LOW);
    }
    else
    {
        Serial.println("Access Denied!");

        digitalWrite(RED_LED, HIGH);
        digitalWrite(GREEN_LED, LOW);
    }
}
```

```

delay(3000);

digitalWrite(RED_LED, LOW);
}
// Stop communication with the card
rfid.PICC_HaltA();
rfid.PCD_StopCrypto1();

delay(500);
}

// Function to compare RFID UID
bool checkUID()
{
  if (rfid.uid.size != sizeof(authorizedUID))
  {
    return false;
  }

  for (byte i = 0; i < rfid.uid.size; i++)
  {
    if (rfid.uid.uidByte[i] != authorizedUID[i])
    {
      return false;
    }
  }

  return true;
}

```

OBSERVATION TABLE

S. No.	RFID Card Used	UID Read	Result (Correct/Wrong)	LED Glowing
1	Card A (registered)	A1 B2 C3 D4	Correct RFID	Green LED ON
2	Card B (unregistered)	5F 6A 7B 8C	Wrong RFID	Red LED ON
3				
4				
5				

RESULT

The RFID-based access control system was successfully designed and implemented using Arduino and the MFRC522 module with only two LEDs. The Green LED turned ON whenever the correct (authorized) RFID card was scanned, and the Red LED turned ON whenever a wrong (unauthorized) RFID card was scanned. The system responded correctly and consistently across multiple trials.

PRECAUTIONS

- Always power the MFRC522 module with 3.3V, never with 5V, to avoid damaging the IC.
- Ensure all SPI connections (SCK, MOSI, MISO, SS) are correctly and firmly connected.
- Keep the RFID tag within 2–3 cm of the reader antenna for reliable reading.
- Avoid placing metal objects near the MFRC522 antenna as they interfere with RF signals.
- Double-check LED polarity (anode/cathode) before connecting to avoid reverse biasing.
- Use current-limiting resistors with LEDs to prevent excess current damage.
- Ensure only one authorized UID is entered correctly in the code (case of hex values matters for comparison).

APPLICATIONS

- Simple door/room access indication systems.
- Attendance marking systems in schools/colleges.
- Secure locker and cabinet access indication.
- Entry verification systems for labs and restricted areas.
- Basic identity-check systems in embedded system learning projects.

CONCLUSION

This experiment demonstrated the practical implementation of a simple RFID-based identification system using an Arduino, an MFRC522 reader, and only two LEDs. The Green LED reliably indicated a correct (authorized) RFID card, while the Red LED indicated a wrong (unauthorized) card, showing the basic working principle of UID-based identification using low-cost, readily available components.